

Spaceship Titanic

Mathematical Optimization Algorithms

Winter Semester 2025/2026, January 2026

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Introduction



- Accuracy: 95% or better
- Trying different:
 - layers
 - neuron counts
 - optimizers
 - hyperparameters (such as learning rate or weight decay)

- training file, 8694 rows, column names:
 - PassengerId
 - HomePlanet
 - CryoSleep
 - Cabin
 - Destination
 - Age
 - VIP
 - RoomService, FoodCourt, ShoppingMall, Spa, VRDeck
 - Name
 - Transported
- test file, 4277 rows, same columns as training but without 'Transported'

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Some values are not numerical



Neural network architecture

- Normalization
- Dropout (in layers)
- Layers:
- Optimizers:
- Activation functions:

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Process

Experimentation

Conclusions/Issues

